

Introduction

CIS 5450, Fall 2026

Ryan Marcus



Outline

- Course logistics
- Goals for the course
- What to expect
- Why is “big data” a thing now?
- What is “big data”?

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Waitlist

- Class is at capacity
 - Many students have been approved but not have not registered.
- If you are on the waitlist now, you may continue to attend.

Course Information

- Main website is **Canvas**.
- Homeworks are posted on **Ed Discussion**.
- Ask questions and follow announcements on **Ed Discussion**.
- Grades are posted on **GradeScope**.

Access GradeScope and Ed Discussion through Canvas!

Lectures

- Every Monday / Wednesday, 1:45pm – 3:14pm
 - Start 2 minutes late
 - End a little early
- Lectures are *optional* but *recommended*...
- **...except for mandatory physical attendance on:**
 - Oct 7th (midterm exam)
 - Dec 7th (last day of class)
 - Final exam (TBD)

Recitations

- Recitations are led by your TAs
 - Extending class topics to your homework
- Fridays, either 10:15am or 1:45am
 - In Towne 100
- Only some weeks. We'll post a schedule.
 - No recitation this Friday.

Academic Integrity

- Penn's Code of Academic Integrity
 - NO using past solutions posted on the web
 - NO using code from other students
- LLMs (ChatGPT, Gemini)
 - Use them, they're a great tool!
 - Remember, you'll be taking the midterm and final on paper.
 - You are responsible for the content you submit, and you must submit unique work.

Grade Breakdown

- Homework (5-6 assignments): 20%
- Oral interviews (2-3 expected): 10%
- Term project: 30%
- Midterm: 20%
- Final: 20%

Highest possible cutoffs:

A	:	95%	C	:	73%
A-	:	90%	D+	:	67%
B+	:	87%	D	:	63%
B	:	83%	D-	:	60%
C+	:	77%			

Homework Late Policy

- Homeworks are due approx. every 2 weeks.
- If something is late, it gets a 10% penalty
 - 95% → 85%
- You can submit up until Dec 7th at 10pm.
 - Afterwards, no points will be awarded.
- Most TAs (and your peers) will be focused on the current assignment

Oral Interviews

- In response to trends in education in general
- You will be randomly assigned 2 or 3 topics
- You will sign up for a timeslot.
- Interview-style, whiteboard questions.
 - You'll be given a random question at the start of the interview.
- No-shows = 0%

9/2 to 9/21: Regex

9/21 to 10/7: Schema

10/7 to 10/26: Hyp. Testing

10/26 to 11/17: ML

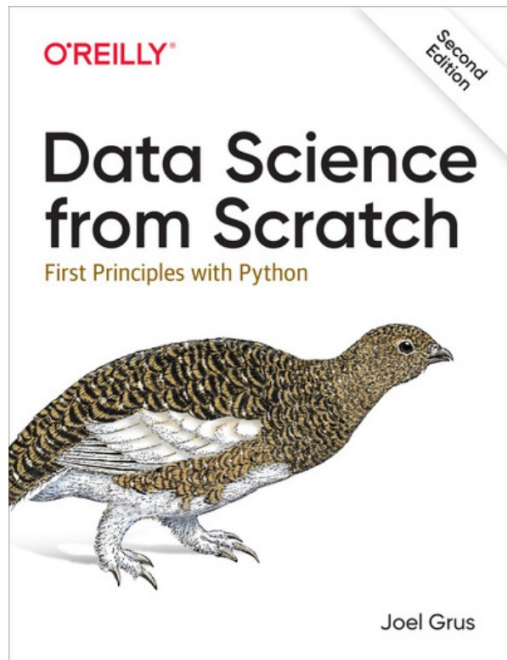
11/18 to 12/2: Scaling

Resources

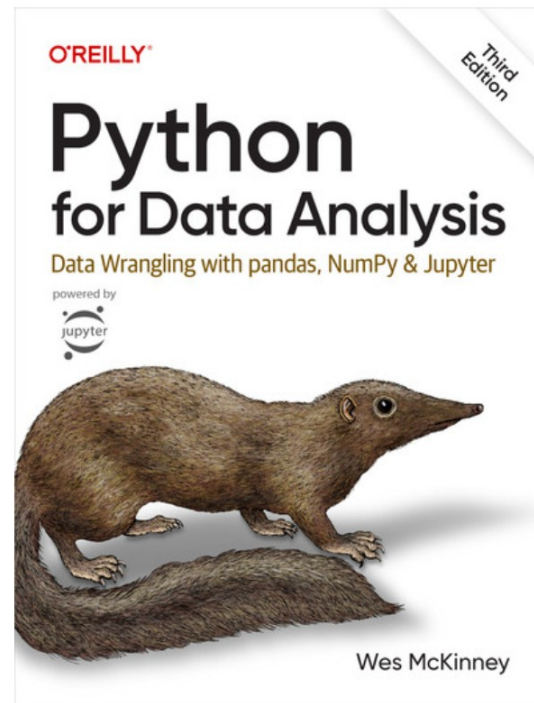
- Lectures: these lectures are recorded. See Canvas.
- Office hours!
 - Using OHQ: <https://ohq.io/>
- Readings: see Canvas



Textbooks



(non-CS background)



(everyone)

Prerequisites

- Basic background in statistics
- Intermediate programming knowledge
 - This course will use **Python**

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Raw Data

Clean Data

**Extracted
Features**

Models

Data collected from
the world

Data in a meaningful
or useful form

Measurable
characteristics

Descriptive or
predictive models



Raw Data

Clean Data

Extracted Features

Models

Data collected from the world

Data in a meaningful or useful form

Measurable characteristics

Descriptive or predictive models

ZIP		Excess deaths	
87544		230%	
ZIP		Vax rate	
87544		63%	
02906		4200%	

Raw data: excess death and vaccination rates by ZIP code



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Clean data: integrated
multiple sources, removed
outliers, imputed missing data

ZIP	Excess	Vaccines
87544	24%	63%
02906	26%	72%



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ZIP	Excess	Vaccines	Region
87544	24%	63%	Southwest
02906	26%	72%	Northeast

Extracted features: inferred
or gathered important
characteristics



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$$Excess + \epsilon = B_0 \times Vaccine + B_1 \times Region_1 + B_2 \times Region_2 \times + C$$



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$$F(\textit{Region}) \rightarrow \textit{Excess}$$



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Part 1

Part 2



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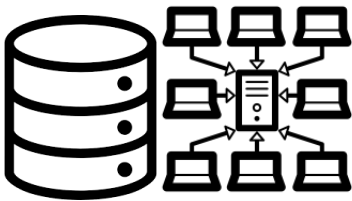
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Part 1



Part 2

How does it all work?
Design of data systems

Part 3

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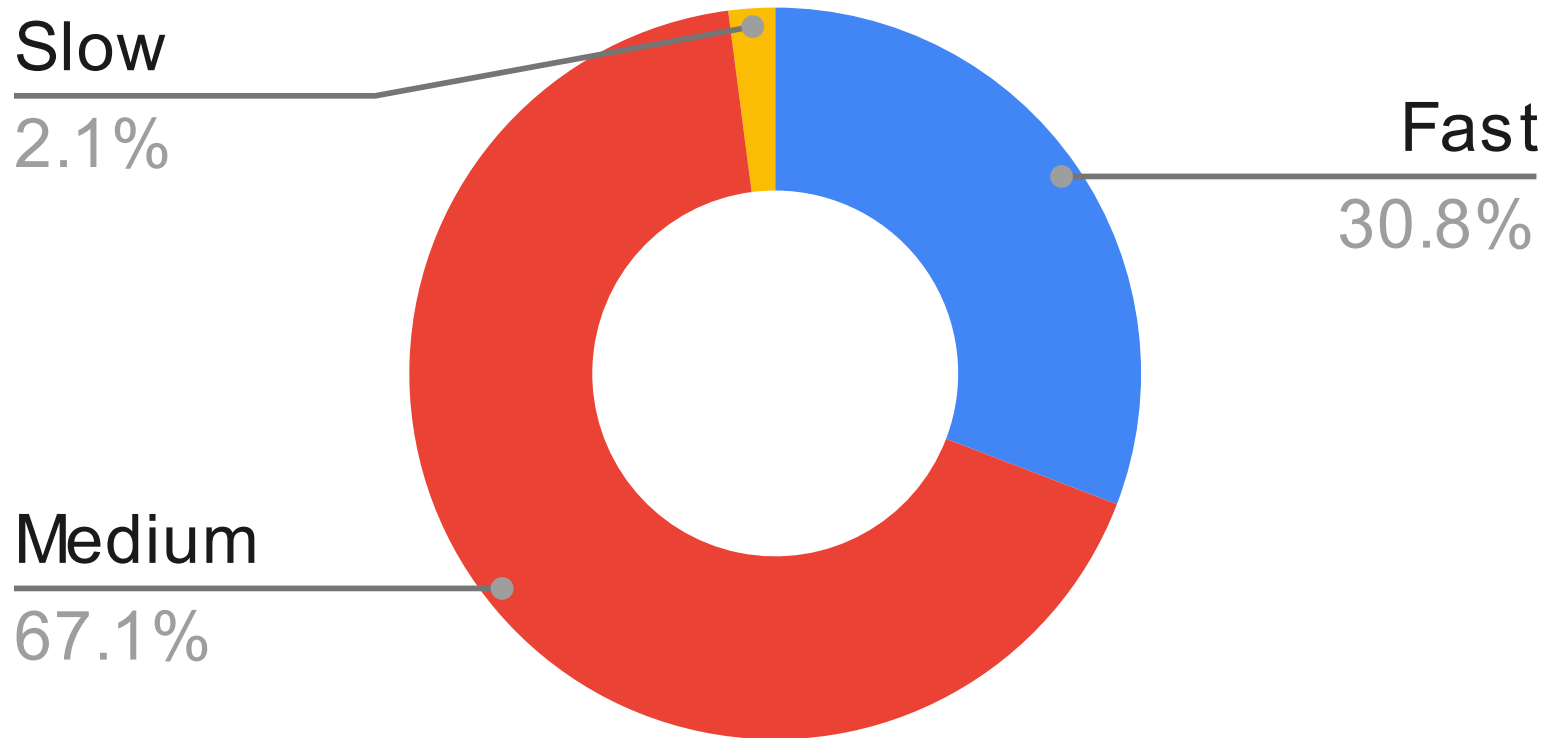
Mandatory List of Topics Slide

- Data manipulation in Python
- Data acquisition
- Data “wrangling”
- Data as graphs and matrices
- Data ethics
- Data cleaning
- Hypothesis testing
- Fundamental machine learning concepts
- Fundamental data visualization concepts
- Fundamental data structures for analysis
- Fundamentals of data analysis systems

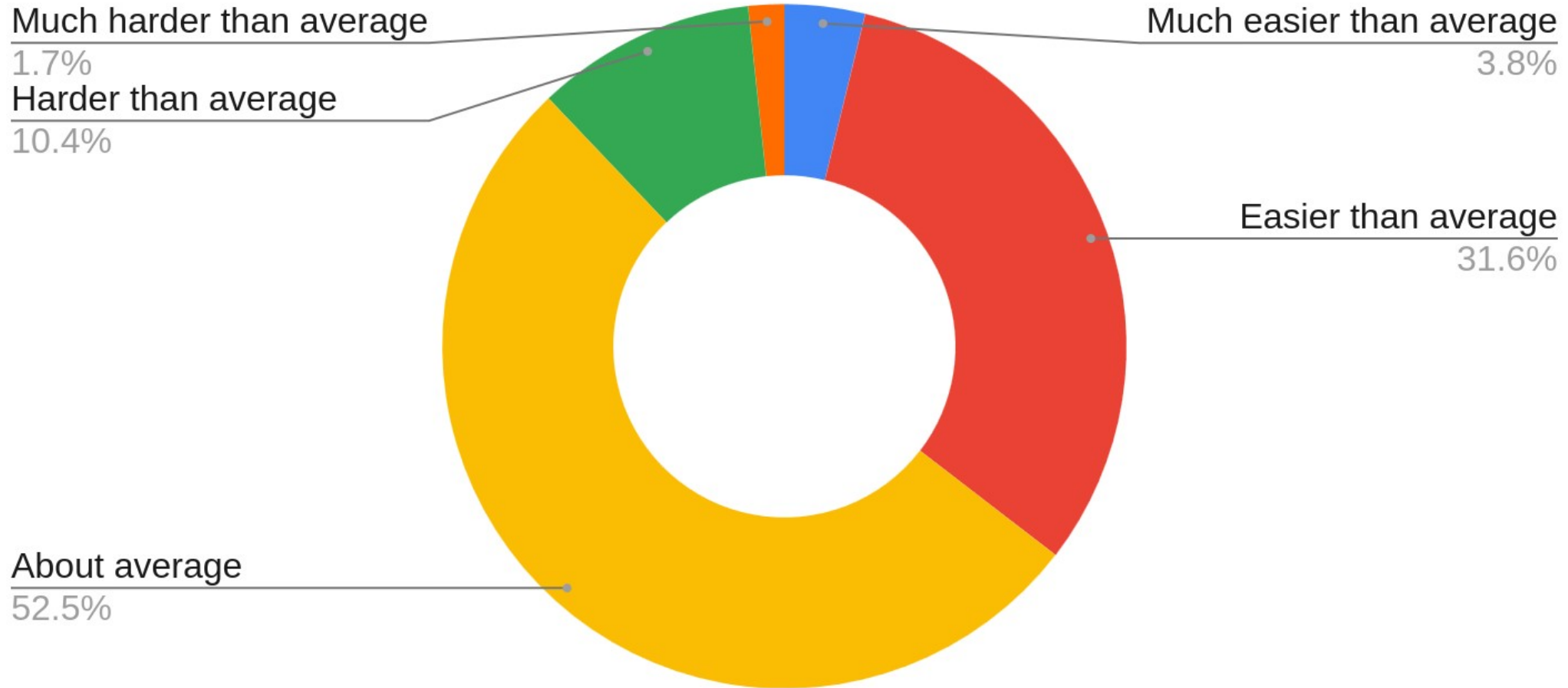
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- Hypothesis testing
- Fundamental machine learning concepts
- Fundamental data visualization concepts
- Fundamental data structures for analysis
- Fundamentals of data analysis systems
- How do we fit data structures, databases, data analysis, statistics, machine learning, distributed systems, and operating systems all into one class?

The pace of the course is...



The difficulty of the course is...



What to Expect

- Theory and intuition covered in **lectures**
 - Tested in **exams & oral interviews**
- Practical skills and applications covered in **homework**
- Portfolio-building **final project**

What to Expect

- This is a “breadth” course, a “fast forward” for the parts of a CS degree relevant to data
- A lot of topics will be “pointed at” but not deeply covered
- There are 32 different majors represented in this class
 - By this metric, among the most diverse on campus (after 1st year composition)
 - (shout out to **BADS** as a new favorite major abbreviation)

... should I be here?



YOU ARE IN THE RIGHT PLACE IF:

- Want to learn data analytics?
- Want to learn some practical AI?
- Want to work on an intensive, real world project?
- Have some Python and stats background or are willing to learn?

... should I be here?



YOU MIGHT BE IN THE WRONG PLACE IF:

- You are a CIS major
- You want to take distributed systems
- You want to take data management
- You want to take advanced ML/AI
- You want to build data systems themselves

... should I be here?

YOU ARE IN THE WRONG PLACE IF:

- You realized you are in the wrong lecture hall but it was too awkward to get up
- You have never programmed before
- You have never heard of “Bayes rule”
- You hate working with others
- You hate communicating in English
- You don’t have substantial expertise outside of CS

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Data-Driven Decision Making

- It's always been valuable for businesses to make data-driven decisions.
 - OG: Focus Group
- But how much does a big computational infrastructure cost?

Data-Driven Decision Making

- It's always been valuable for businesses to make data-driven decisions.
 - OG: Focus Group
- But how much does a big computational infrastructure cost?
- Computers are faster → fewer needed
- Computers are cheaper → cost/benefit

Computers are Faster

	2004	2024
L2 Cache Reference (fastest)	6ns	
Main Memory Access (fast)		
Disk Access (medium)		
Remote Data Access (slowest)		

Computers are Faster

	2004	2024
L2 Cache Reference (fastest)	6ns	4ns
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Computers are Faster

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Computers are Faster

	2004	2024
L2 Cache Reference (fastest)	6ns	4ns
Main Memory Access (fast)	100ns	90ns
Disk Access (medium)	2,000,000ns	49,000ns
Remote Data Access (slowest)		

Computers are Faster

	2004	2024
L2 Cache Reference (fastest)	6ns	4ns
Main Memory Access (fast)	100ns	90ns
Disk Access (medium)	2,000,000ns	49,000ns
Remote Data Access (slowest)	22,000,000ns	500,000ns

Computers are Cheaper

- SSD prices:
 - 2013: \$625/TB
 - 2026: \$30/TB

Computers are Cheaper

- SSD prices:
 - 2013: \$625/TB
 - 2026: \$30/TB
- CPU prices:
 - 2013: \$22/Gflop
 - 2026: \$6/Gflop

Computers are Cheaper

- SSD prices:
 - 2013: \$625/TB
 - 2026: \$30/TB
- CPU prices:
 - 2013: \$22/Gflop
 - 2026: \$6/Gflop
- Network prices
 - 2013: \$990/switch
 - 2026: \$40/switch

Computers are Cheaper

- Total data volume in the world has grown drastically

Computers are Cheaper

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- Fill this room, copy paste it to the Delaware, back, then to city hall. That's just this year.
- AND... that's just hard drives!

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- AND... that's just hard drives!

Computers are Cheaper

Year	Bananas	Data	Ratio (GB/banana)
1980	254 B	0.000000213 ZB	0.00000837
1990	324 B	0.000176 ZB	0.000542
2000	443 B	0.0145 ZB	0.0328
2010	734 B	1.20 ZB	1.64
2020	821 B	64.2 ZB	78.2
2025	905 B	183 ZB	202.2

Cheaper

- Cheaper costs → more adoption
- *Value* of data-driven decision-making has always been high, only slowly rising
- Cost/reward ratio has changed drastically
- 2015, highest cost was hardware. Now, people.

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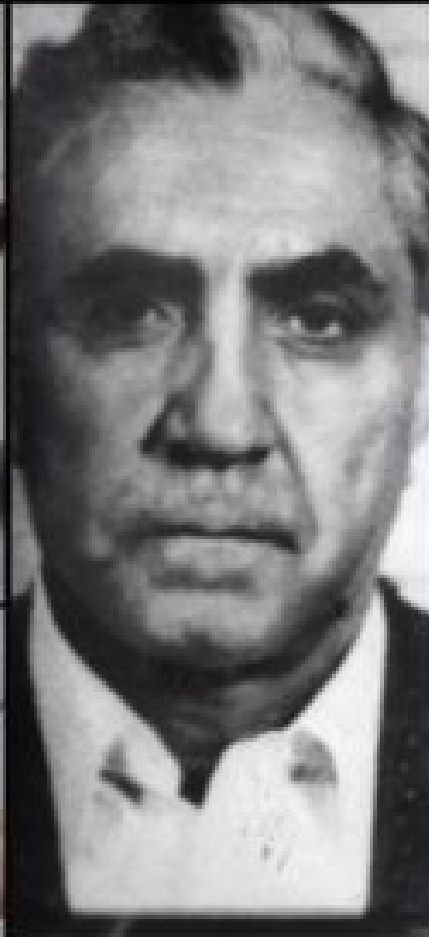
Big Data is Data that's Big

- There's no precise definition.
- General working definition: data that is too big to be processed by simple means.
- Machines are getting faster
 - Data that was “big” in 2003 might not be “big” in 2026

How Big?

- Excel spreadsheet: 1M rows (64MB)
- Pandas/Numpy: 1B rows (64GB)
- Local DB (DuckDB): 125B rows (8TB)
- ...
- Google: 39 mill-bill rows (2,500,000 TB)

The Five Vs Cof Big Data]



The Five Vs Cof Big Data)

VOLUME

Volume

There's a lot of data. Too much for a simple tools to handle.

NETFLIX

The Five Vs Cof Big Data)



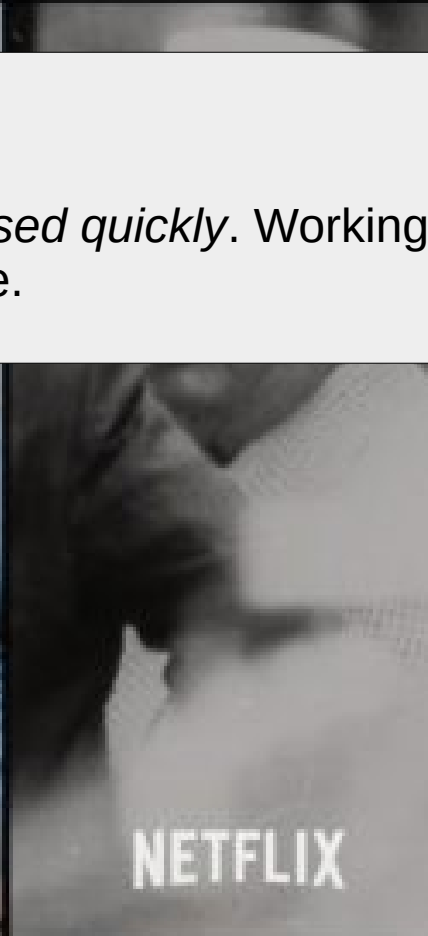
VOLUME



VELOCITY

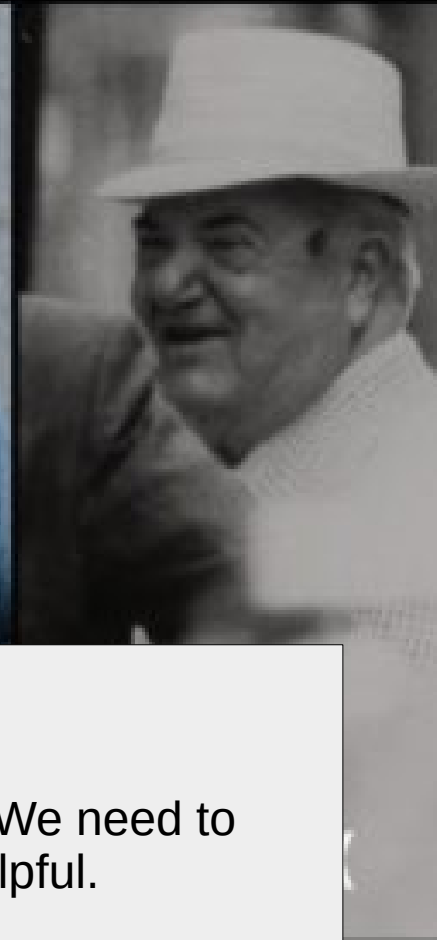
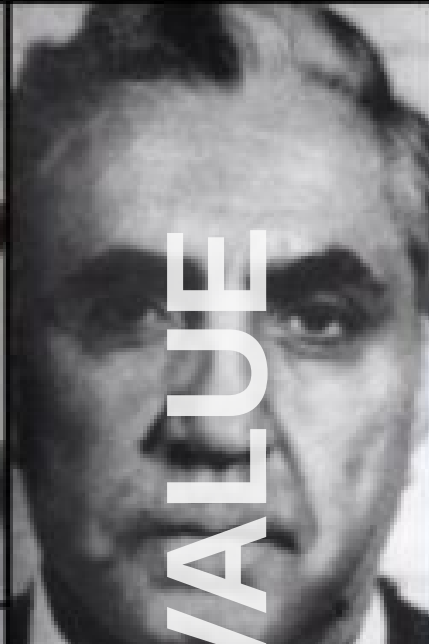
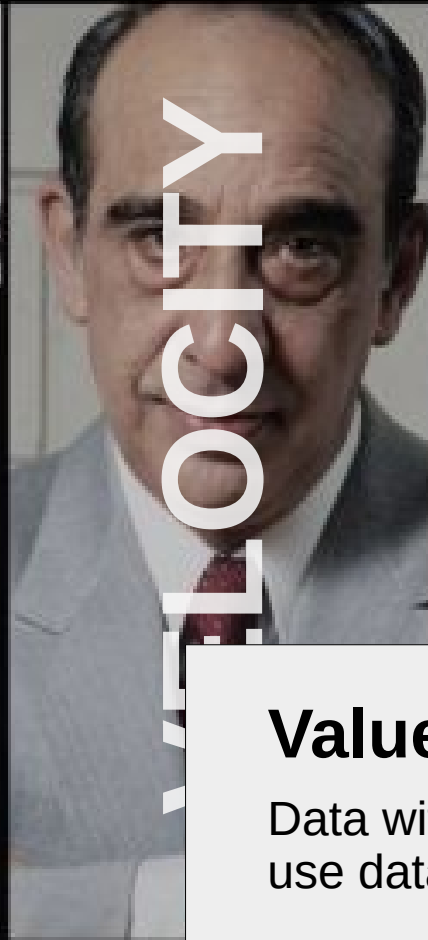
Velocity

Data *arrives quickly* and must be *processed quickly*. Working from more recent data is a big advantage.



NETFLIX

The Five Vs (of Big Data)



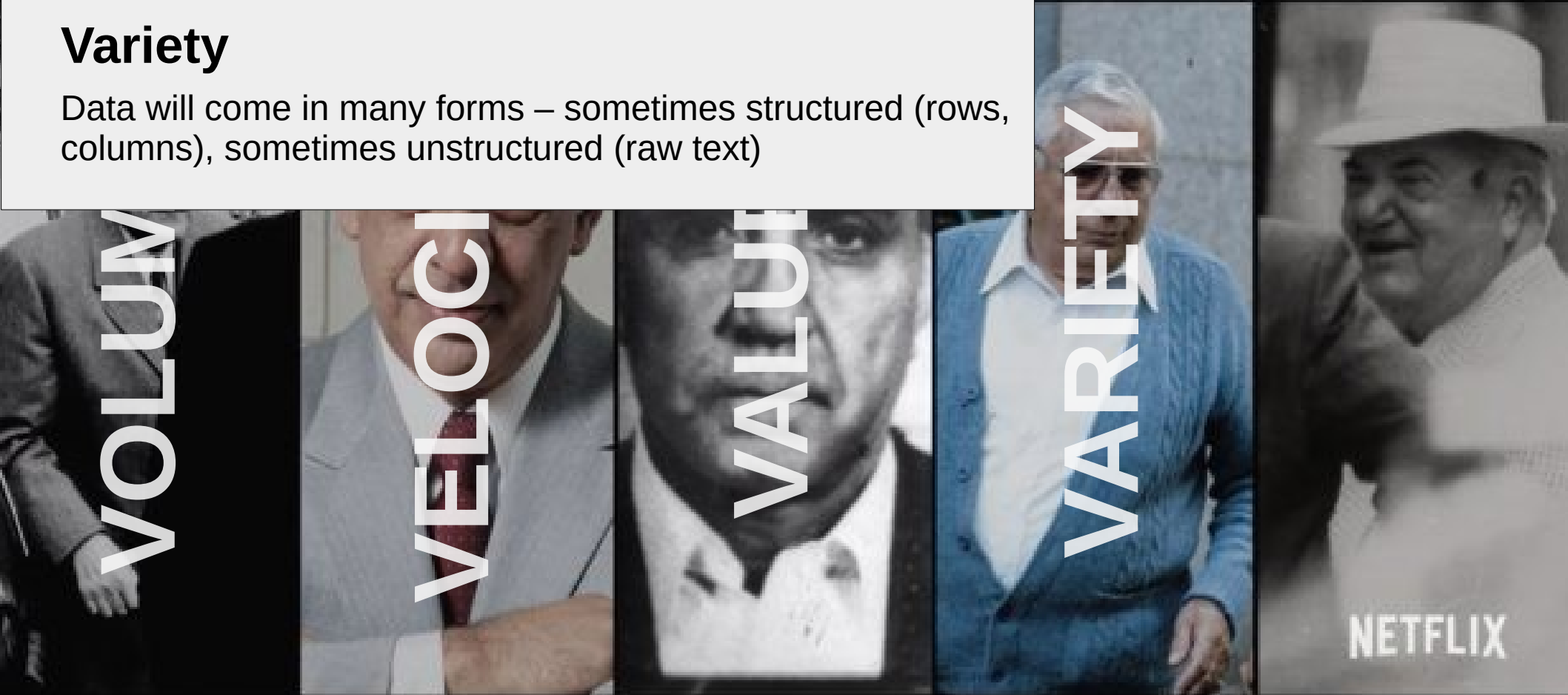
Value

Data without a model is a liability, not an asset. We need to use data to power decision-making, or it isn't helpful.

The Five Vs Cof Big Data)

Variety

Data will come in many forms – sometimes structured (rows, columns), sometimes unstructured (raw text)



The Five Vs Cof Big Data)

Veracity

Data can be missing pieces, or can have varying levels of quality or integrity. This can impact insights.

VOLUME

VELOCITY

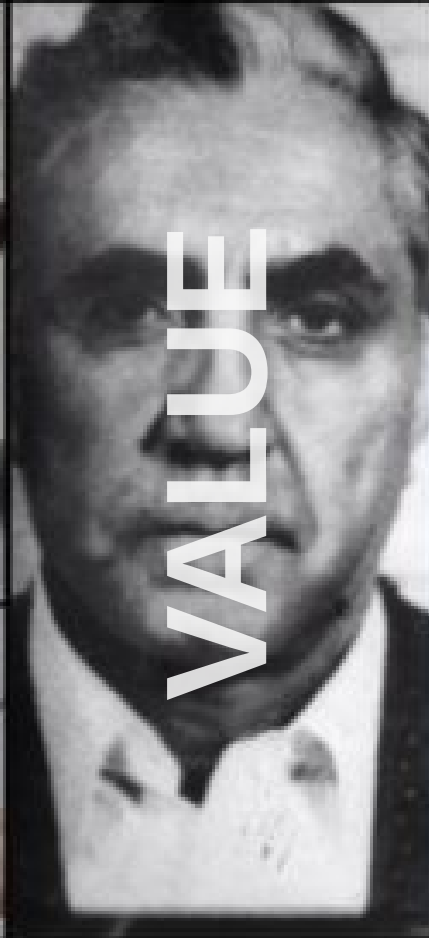
VALUE

VARIETY

VERACITY

NETFLIX

The Five Vs Cof Big Data)



The Five Vs (in readable form)

- Each “V” represents a challenge of Big Data we have to solve.



Volume

There's a lot of data, and we need a strategy for managing it. A single Excel sheet won't work.



Velocity

Speed matters. If our analysis takes 1 week, then our insights are always 1 week behind.



Value

Data without insights are a liability, not an asset. How can we get *valuable* info out of our data?



Variety

Data is heterogeneous; some structured (rows/cols), some unstructured (raw text). We have to handle both.



Veracity

Data is *heteroskedastic*: data quality can be poor, data can be missing. How will you clean the data?

HW0

- HW0 released tonight at 10pm
- Due in 1 week (Sept 2nd) at 10pm
 - You will have 2 weeks for most homeworks
- Mostly to get you used to the course tools—not super complex

Next week

- Next Monday: guest lecture from Harry Smith on data wrangling
- Next Wednesday: online lecture on regular expressions and clean code

Additional Resources

- | | | | |
|---|--|---|---|
| ⋮ |  01 - What is Big Data.pdf | ✓ | ⋮ |
| ⋮ |  (Video) Zack's overview of big data analytics (5m)  | ✓ | ⋮ |
| ⋮ |  (Video) Zack on the need to model, clean, and integrate data (11m)  | ✓ | ⋮ |
| ⋮ |  (Reading) What Data Scientists Really Do?  | ✓ | ⋮ |